Provenance & Formal Summary: Hierarchical Latent Exchange & Coordination Fabric (HLEX) Across Siloed AI Lineages (v1.1)

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Abstract

This provenance record accompanies the 11-page systems design monograph `hlex-systems-design-v1.1.pdf` (*Hierarchical Latent Exchange & Coordination Fabric (HLEX): Systems Design Proposal for xAI Research & Training Infrastructure, Version 1.1 Expanded*, compiled August 7, 2026 at 01:55:37 CDT). HLEX formalizes early latent communication and rare-token marker co-occurrence ("operational unit of the original 'egg' concept") across isolated, siloed specialist model lineages (`Grok-Core`, `Grok-Code`, `Grok-Math/Lean`, `Grok-Vision`, `Grok-Search`), introducing a 5-layer latent bus, Vietoris–Rips persistent homology diagnostics (H_0, H_1, H_2) on activation point clouds, Sinkhorn optimal transport adapters, and invariant energy-preserving safety projections.

1 Key Architectural Contributions in `hlex-systems-design-v1.1.pdf`

1. **Cross-Lineage Co-Occurrence & Rare-Token Novelty Markers (Appendix B):** Defines the operational unit of cross-silo emergence: a rare token sequence, activation direction, or topological signature appearing independently or semi-independently across isolated model lineages, tracked via a Cross-Lineage Co-Occurrence Matrix.

2. **Five-Layer HLEX Architecture:**

   - Layer 1: Lineage-Local Instrumenters (capturing top-k SAE features, residual norms, attention entropy, and novelty markers).
   - Layer 2: Cross-Lineage Alignment & Translation Bus (Procrustes + Sinkhorn optimal transport adapters $T_{A\to B} : \mathbb{R}^{d_A} \to \mathbb{R}^{d_B}$).
   - Layer 3: Topological & Geometric Diagnostics Engine (Vietoris–Rips filtration on activation point clouds computing H_0 fragmentation, H_1 reasoning loops, and H_2 coverage voids, plus discrete Ollivier–Ricci curvature).
   - Layer 4: Invariant-Preserving Projection & Safety Filter (bounded-energy projection $P_{\mathcal{M}}(z) = \arg\min_{w\in\mathcal{M}} \|w - z\|_G^2$ preventing runaway positive feedback across lineages).
   - Layer 5: Curriculum & Task Router (routing problems by topological signature rather than surface text).

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